Buried, Recovered, Lost Again? The Romanovs May Never Rest

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DNA studies in the 1990s appeared to prove that the remains of the last Russian tsar and his family had been found; a new analysis raises questions

In the summer of 1991, the remains of nine people were unearthed from a shallow grave in central Russia. Forensic experts concluded that the skeletons likely were those of the last tsar, Nicholas II, the tsarina, and three of their five children, whose bodies disappeared after they were shot by the Bolsheviks in July 1918. DNA studies in the mid-1990s supported that claim, and in 1998, a special government panel affirmed the bones to be those of the Romanovs, Russia's ill-starred royal family, along with their doctor and three servants.

A new study, however, challenges this verdict. In the current issue of the *Annals of Human Biology*, a team led by molecular systematist Alec Knight of Stanford University resurrects questions about the discovery of the remains and mounts a blistering attack on the original DNA analysis, contending that the results were tainted. Knight's group also performed the first analysis of the remains of the Grand Duchess Elisabeth, the tsarina's sister. All told, says Knight, "the evidence does not support the claim that the remains are those of the Romanov family."

"That's nonsense," fires back Pavel Ivanov, a molecular biologist at the Engelhardt Institute of Molecular Biology in Moscow who carried out the original DNA studies with Peter Gill of the U.K. government's Forensic Science Service and several colleagues. Ivanov and Gill contend that the new claims are flawed.

The new findings "do not add to the slim doubts about the remains. But they do not take away from the doubts either," says Evgeny Rogaev, a molecular geneticist at the Center of Mental Health in Moscow and the University of Massachusetts Medical School in Worcester, who was invited by Romanov descendants and the Russian government to conduct an independent examination.

The debate could create a stir in Russia. The Russian Expert Commission Abroad, an expatriate group that doubts the authenticity of the remains, is calling for a new examination, claiming support from the Russian Orthodox Church. The plea, however, may fall on deaf ears: "The case is closed," says a Putin Administration official.

The lingering uncertainty stems in part from the intrigue surrounding the demise of the Romanovs, said to have been killed and disposed of near Ekaterinburg. The discovery of a communal grave in the vicinity was reported in 1989. Two years later Russia's chief forensic medical examiner unearthed nine badly damaged skeletons and, after a series of forensic tests, came up with a tentative ID.

The Gill and Ivanov team amplified short tandem repeats from DNA, which confirmed that the purported tsar, tsarina, and three girls belonged to the same family.

Vanished royalty. Tsar Nicholas II and family in one of their last official photographs before they disappeared.

If correct, the bodies of one daughter and the tsarevich, Alexei, were missing. More curious was an analysis of mitochondrial DNA (mtDNA), which is passed from mother to child. The team compared mtDNA fragments from the nine skeletons to blood samples from the Duke of Edinburgh, Prince Philip—a grandnephew of the tsarina—and two living descendants of the tsar's maternal grandmother. The tsarina and the children matched Prince Philip, but the tsar's mtDNA was heteroplasmic—having cytosine and thymine at one site. The relatives had thymine or cytosine. Despite that apparent mismatch, the group reported in the February 1994 issue of *Nature Genetics* that the odds of the remains *not* being those of the Romanovs were no less than 700 to 1.

Two years later, Ivanov, working with a U.S. team, provided what appeared to be the clincher: The mtDNA of the remains of the Grand Duke Georgij Romanov, the 28-year-old brother of the tsar who died of tuberculosis in 1899, showed the same heteroplasmy. Rogaev in 1997 and 1998 sequenced DNA from the femur presumed to be from Nicholas II and from the blood of a nephew of Nicholas II. In findings submitted to the Russian commission, he concluded that the DNA matched.

Knight and colleagues contend, based on recent historical research, that the discovery and removal of the remains is "characterized by extreme irregularities at every level," and that "crucial evidence has been proven fraudulent." The most damning shortcoming, they charge, is that samples of old DNA must have been contaminated with "fresh" DNA that skewed the analysis. They argue that a sequence of 1123 base pairs, for example, was too long to have come from old bones.

The Knight team contributes some new data: a DNA analysis of the Grand Duchess Elisabeth's shriveled finger. The sequence did not match the reported sequence of the tsarina. To Knight's group, this means that "it is probable that the Ekaterinburg remains were misidentified." But the sequence didn't match Prince Philip's mtDNA either, so it might not have come from Grand Duchess Elisabeth, says Ivanov. Knight's team acknowledges that it may have been from a contaminant.

The Knight paper is "intriguing," says Anne Stone, an expert on ancient DNA at Arizona State University in Tempe, who sequenced the purported remains of the Wild West outlaw Jesse James. Notwithstanding the Elisabeth–Prince Philip puzzle, she says, the "the ancient DNA work looks like it was performed according to today's standards and looks good."

Gill is not impressed, insisting that his team's forensic DNA work "set the standard." The new paper so completely mischaracterizes his work, he says, that it "comes across as vindictive and political."

Rogaev says that further DNA studies would be worthwhile. Knight agrees; one possible way to settle the identity of the tsarina, he says, is to invite other living descendants of Queen Victoria, her maternal grandmother, to donate blood for mtDNA sequencing. Ivanov, meanwhile, contends that the scientific argument is too weak to warrant debate. And what about the Romanovs? They must be turning in their graves.

–RICHARD STONE

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